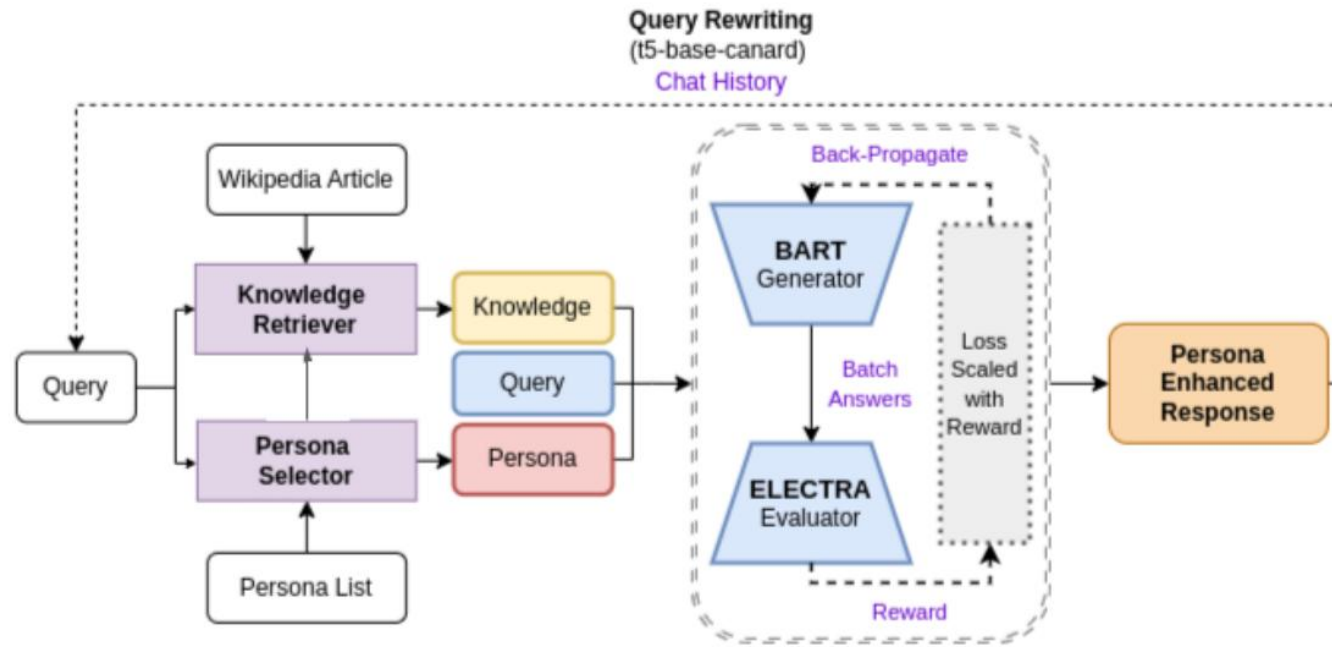


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Persona based Proactive Conversational Agent

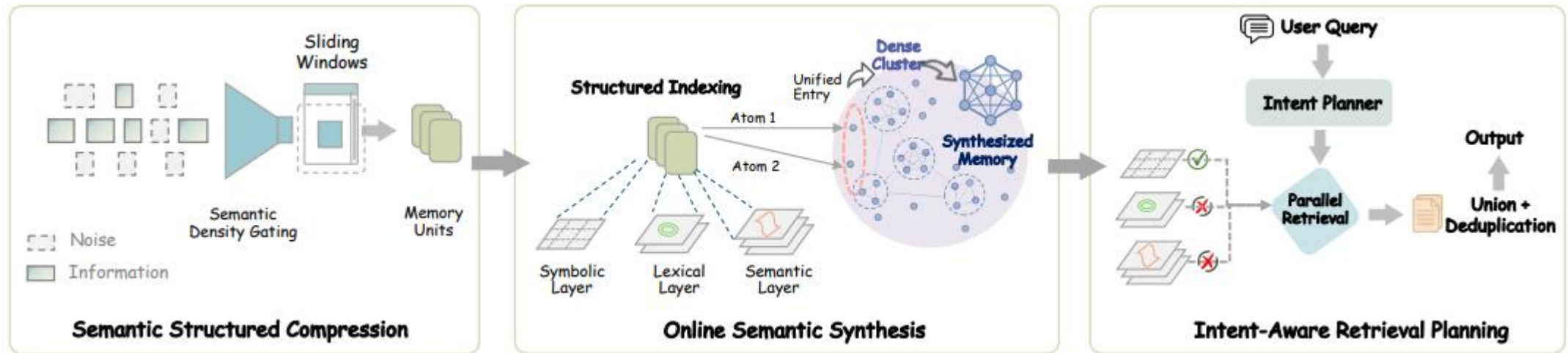
Personalized Response Generation using Dynamic Knowledge Retrieval and Persona Adaptive Queries



1. The model employs **Dynamic Knowledge Retrieval** to supplement responses with information from background sources, using Dense Passage Retrieval (similar to RAG).
2. A dedicated **Persona Selector** module chooses a persona that aligns with the user's query. This is modeled as a multi-label classification task, which includes an option for "no-persona" for generic queries.

1. **Query Rewriting:** Utilizes a LLM model to refine the user's input based on chat history.
2. **Response Generation:** A **BART** generator is paired with an **ELECTRA** evaluator.
3. **Reward Function:** The system uses a balancing reward function that considers generative capabilities like coherence to ground truth responses and persona alignment.

SimpleMem: Efficient Lifelong Memory for LLM Agents



1. **Semantic structured compression:** This stage actively filters redundant interaction content, such as social chit-chat or repetitive logs, using an entropy-based filtering. Temporal dialogue data are split into overlapping windows. Raw dialogue windows are evaluated based on the introduction of new entities and semantic novelty.
2. **Online Semantic Synthesis:** It is a recursive memory consolidation step. Units are grouped into clusters based on both semantic similarity and temporal proximity.
3. **Intent-aware retrieval:** This mechanism dynamically adjusts the retrieval scope based on the estimated complexity of the user query. A lightweight classifier determines if a query requires a simple fact lookup or complex multi-step reasoning. Retrieval depth (k) is modulated accordingly, ranging from a minimum of 3 entries for simple queries to a maximum of 20 for complex ones, ensuring a balance between accuracy and token efficiency. This is done using a reasoning LLM.

Target-Guided Conversations

1. The goal of target-guided dialogues is to achieve the designated target effectively and efficiently through engaging conversations
2. **Topic-shift Detection:** Discover the topic drift in the user utterances during the conversation. Fine-tune sentence classifier to classify the utterances into major, minor and off topics for detecting the digression from the target topic of the conversation.
3. **Topic Planning:** Enables the conversation to follow an expected topic direction. Can leverage external knowledge graphs for improving the quality of topic transitions with graph reasoning techniques. -> *GraphRAG retrieval technique*
4. **Topic-guided Response Generation:** Based on the topic/keyword selected by the planner, augment the user query to generate topic related response

Profile Building

1. Objective is to build a persona profile from user dynamically as conversation/ journalling continues
2. **Medical Persona Ontology :** Specify keys/ set of features we want to track. Eg: sleep quality, diet pattern, exercise frequency
3. **Persona-Extractor:** Can be LLM based methods to extract health-related facts into predefined schema or rule based like Named Entity Recognition
4. **Continuous Updates:** As conversation moves, keep on updating the profile. Store statistics to capture trends, this can be later used for task planning
5. **Use Persona for Consistency:** This profile can also be used to create topic planning.